

Luca Lazzaroni

Curriculum Vitae

Personal data and education

Luca Lazzaroni was born in Pietra Ligure on August 6, 1995. Currently, he resides in Genova (GE).

In 2014, he obtained the **Scientific High School Diploma** at the "Arturo Issel" State Scientific High School in Finale Ligure (SV) with a final grade of 100/100.

On 27 October 2017, he graduated in **Electronic Engineering and Information Technologies** (class L-8 ex D.M. 270/2004) at the University of Genoa with a thesis entitled "Model-based design of Lego robot control in Simulink environment".

On 24 July 2020, at the same University, he obtained the **Master's Degree in Electronic Engineering** (class LM-29 ex D.M. 270/2004) with a thesis entitled "Design and implementation of a multiplatform execution engine for IoT applications" with a final grade of 110/110.

Subsequently, he won a PhD scholarship at the Department of Naval, Electrical, Electronic and Telecommunications Engineering (DITEN) of the University of Genoa, entitled "Deep learning on the edge", which he used from 1 November 2020 to 31 October 2023 (XXXVI cycle). He passed, with a positive outcome, the exam for the achievement of the title of **PhD in Science and Technology for Electronic and Telecommunications Engineering** on 14/03/2024, presenting a final dissertation entitled: "The Facets of Edge AI in Automotive: Exploring Embedded Frameworks, Voice Assistants, and Deep Reinforcement Learning".

Subsequently, he was awarded a **Research Grant** at the Department of Naval, Electrical, Electronic and Telecommunications Engineering (DITEN) of the University of Genoa, entitled "Machine Learning models for the processing of data from autonomous vehicles", which he has used from 1 November 2023 to 28 February 2025.

Since 1 March 2025, he is **Assistant Professor** (RTD-A) at DITEN, in the scientific-disciplinary sector IINF-01/A - Electronics.

In the second session of the year 2020 he passed the **State Examination** for the qualification to practice the profession of Information Engineer.

He is author/co-author of **67 scientific publications** in international journals (17), international (44) and national (6) conferences. The complete list of publications resulting from the research activity is reported in the "Publications" section.

Editorial curation

- He served as *Program Chair* and *Editor* at the 2025 Applications in Electronics Pervading Industry, Environment and Society, currently in press.
- Since December 18, 2025, he has been part of the Editorial Board of the journal "Sensors", as an *Early Career Editorial Board Member* (<https://www.mdpi.com/news/14234>).
- From 10 April 2024 to 25 August 2025, he has edited, as *Guest Editor*, the Special Issue "Tiny Machine Learning for Energy Applications" in the international journal "Energies", dedicated to the application of artificial intelligence techniques on sustainable embedded devices (https://www.mdpi.com/journal/energies/special_issues/4Q944YV2F8).

Contributions to the organization of scientific conferences

- He is *Program Chair* of the international conference "International Conference on Applications in Electronics Pervading Industry, Environment and Society", 2025 edition (<https://applepies.unige.it/organization>)
- He held the role of *Chairman* in the "1st Short Oral session" of the international conference "International Conference on Applications in Electronics Pervading Industry, Environment and Society", Genoa, Italy, 28-29 September 2023.
- He is a member of the *Program Committee* of the international conference "Games and Learning Alliance Conference" since 2023.
- He is a member of the *Program Committee* of the international conference "International Conference on Applications in Electronics Pervading Industry, Environment and Society" since 2023.

Reviewer Activity

He has been a reviewer for the following international scientific journals:

- *Engineering Applications of Artificial Intelligence* (since 2024).
- *IEEE Access* (since 2023).
- *Journal of Systems Architecture* (since 2023).
- *Sensors* (since 2024).
- *IET Intelligent Transport Systems* (since 2025).
- *IEEE Transactions on Intelligent Vehicles* (since 2026).

He has been a reviewer for the following international conferences:

- *2026 IEEE International Conference on Robotics & Automation (ICRA)*.

- *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (2025 edition).
- *Games and Learning Alliance* (2023, 2024 editions).
- *International Conference on Applications in Electronics Pervading Industry, Environment and Society* (2021, 2022, 2023, 2024 editions).

Participation as a speaker at scientific conferences

He has participated as a speaker (making oral presentations) in the following international conferences and meetings:

- *2025 International Conference on Applications in Electronics Pervading Industry, Environment and Society*, Torino (TO), Italy, 11-12 September 2025.
- *43rd IEEE International Conference on Consumer Electronics (ICCE)*, Las Vegas (NV), USA, 11-14 January 2025.
- *2024 International Conference on Applications in Electronics Pervading Industry, Environment and Society*, Torino (TO), Italy, 19-20 September 2024.
- *55th Annual Meeting of the Associazione Società Italiana di Elettronica (SIE)*, Genoa (GE), Italy, 24-26 June 2024.
- *2023 International Conference on Applications in Electronics Pervading Industry, Environment and Society*, Genoa (GE), Italy, September 28-29, 2023.
- *54th Annual Meeting of the Associazione Società Italiana di Elettronica (SIE)*, Noto (RG), Italy, September 7-9, 2023.
- *2022 International Conference on Applications in Electronics Pervading Industry, Environment and Society*, Genoa (GE), Italy, 26-27 September 2022.
- *53rd Annual Meeting of the Associazione Società Italiana di Elettronica (SIE)*, Pizzo (VV), Italy, 2022, 7-9 September 2022.
- *2021 Games and Learning Alliance Conference*, La Spezia (SP) (online), Italy, December 1-2, 2020.
- *2021 International Conference on Applications in Electronics Pervading Industry, Environment and Society*, Pisa (PI), Italy, 21-22 September 2021.
- *52nd Annual Meeting of the Associazione Società Italiana di Elettronica (SIE)*, Trieste (TS) (online), Italy, July 7-9, 2021.
- *2020 Games and Learning Alliance Conference*, Laval (online), France, December 9-10, 2020.
- *2020 International Conference on Applications in Electronics Pervading Industry, Environment and Society*, online, 19-20 November 2020.

Teaching activity

- He teaches the following courses at the University of Genoa:
 - “Distributed Electronic Systems and Technologies for Environmental Monitoring” (code 94631), A.Y. 2024/2025, 2025/2026.
 - “Tiny Machine Learning” (code 114752), A.Y. 2024/2025, 2025/2026.
- He is co-supervisor of the following Bachelor's and Master's theses at the University of Genoa:
 - 10 Master's Degree Thesis in Electronic Engineering (LM-29).
 - 12 degree theses in Electronic Engineering and Information Technologies (L-8).
 - 2 Master's Degree theses in Computer Science (LM-18).
 - 7 Master's Degree Thesis in Technological Engineering for Strategy (and Security) (LM/DS).
- He is a Subject Expert for the following courses at the University of Genoa (from A.Y. 2022/2023):
 - “Machine Learning for Automated Driving” (code 108716).
- He has been an adjunct professor for the following courses at the University of Genoa:
 - “Tiny Machine Learning” (code 114752), A.Y. 2024/2025.
 - “Computer Graphics Arch.- Games And Simulation” (code 172317), A.Y. 2023/2024.
- He has carried out teaching support activities for the following courses at the University of Genoa:
 - “Autonomous Agents in Games” (code 198216), A.A. 2022/2023.
 - "Fundamentals of Programming of Electronic Systems" (code 166061), A.Y. 2020/2021.
 - Adoption of innovative teaching methodologies for the courses "Fundamentals of Electronic Systems Programming" (code 166061), "Makers Approach to Electronic Design" (code 186985), A.Y. 2020/2021.

Academic appointments

- He is a member of the Admission Committee for the Master's Degree Course in Engineering for Natural Risk Management

Research projects

He has participated in the following research projects:

- **European project** HORIZON-CL5-2023-D6-01-02 "Generation of scenarios for development, training, virtual testing and validation of CCAM systems" (Acronym: **Synergies**), as a member of the research unit of the University of Genoa. The project sees the collaboration between the most important European car manufacturers and many universities active in the sector. Synergies aims to generate a wide range of test scenarios for training, testing and validation of CCAM systems, with a focus on urban and rural traffic, for which there is significantly less knowledge about relevant scenarios than highway driving.
- **European project** H2020 (GA 101006664) "Addressing challenges toward the deployment of higher automation" (Acronym: **Hi-Drive**), as a member of the research unit of the University of Genoa. The project sees the collaboration between the most important European car manufacturers and many universities active in the sector. Hi-Drive develops and tests advanced automated driving capabilities on a large scale, moving from conditional automation (SAE L3) to high automation, in different scenarios and road contexts.
- **National project** PRIN2022 "Green Machine Learning for the IoT" (Acronym: **GEMINI**), as a member of the research unit of the University of Genoa. The project is in collaboration between DII - UNITN and DITEN - UNIGE. GEMINI aims to implement a new framework that supports the development of fully sustainable TinyML applications deployed along an edge-to-cloud continuum. The GEMINI framework is intended for battery-free embedded devices, including those without network connectivity.
- **National project** POR FESR Piedmont 2014-2020 "ADINA: Integrated Digital Assistant Automotive" (Acronym: **ADINA**), as a member of the research unit of the University of Genoa. The project is in collaboration with Lifetouch, an Italian company specializing in the automotive sector. ADINA aims to develop a Virtual Assistant integrated on specific hardware to be implemented on the car and connect to the vehicle's systems. The virtual assistant uses artificial intelligence algorithms to understand the user's requests and then act on the vehicle and infotainment systems.

Membership in professional associations

- He is a member of the international scientific association IEEE with the rank of *IEEE Member*.
- He is a member of the Italian Society of Electronics Association (SIE) within the Research Unit of the University of Genoa (since 2021), pertaining to the research area "Electronic systems and applications".

Awards and recognitions

- He was the winner of the "Best Presentation of the Session" award for the "Edge Computing" session at the *43rd IEEE International Conference on Consumer Electronics (ICCE)*, held in Las Vegas, NV, USA, on January 11-14 2025 with the following contribution:

- A. Dabbous, L. Lazzaroni, F. Sakr, R. Berta, M. Fresta, and F. Bellotti, "Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices".

Publications

International Journals

1. L. Lazzaroni, R. Berta, V. Soltanmuradov, D. Martín Gómez, and F. Bellotti, "StabilEdge, Formal Verification of Edge AI Applications," *IEEE Access*, vol. 14, pp. 37416–37434, 2026, doi: 10.1109/ACCESS.2026.3667991.
2. H. Rojhan, F. Bellotti, M. Fresta, L. Lazzaroni, and R. Berta, "Usability, Technicality, Interoperability (UTI), a Comprehensive Assessment Framework for Digital Platforms," *IEEE Access*, vol. 14, pp. 44886–44903, 2026, doi: 10.1109/ACCESS.2026.3676567.
3. L. Lazzaroni, F. Bellotti, A. Pighetti, L. Forneris, M. Fresta, H. Ballout, C. Oh, S. Pantawane, and R. Berta, "An Embedded System Architecture for Robust Deep Learning-Based Traffic Light Detection," *IEEE Sensors Journal*, vol. 25, no. 24, pp. 44859–44874, Dec. 2025, doi: 10.1109/JSEN.2025.3627477.
4. L. Forneris, R. Berta, M. Fresta, L. Lazzaroni, H. Rojhan, C. Oh, A. Pighetti, H. Ballout, F. Tango, and F. Bellotti, "A Deployment-Oriented Simulation Framework for Deep Learning-Based Lane Change Prediction," *IEEE Signal Processing Letters*, vol. 33, pp. 136–140, 2026, doi: 10.1109/LSP.2025.3638676.
5. A. Pighetti, F. Bellotti, R. Berta, A. Cavallaro, L. Lazzaroni, and C. Oh, "RAFT: Regularized Adversarial Fine-Tuning to Enhance Deep Reinforcement Learning for Self-Parking," *IEEE Sensors Letters*, pp. 1–4, 2025, doi: 10.1109/LSENS.2025.3600982.
6. M. Fresta, F. Bellotti, L. Lazzaroni, H. Ballout, A. Pighetti, Ó. Ameneiro-Prieto, M. Porta-Lorenzo, D. Sánchez-Louzán, C. González-Escudeiro, F. Tango, and R. Berta, "On-Board Deployability of a Deep Learning-based System for Distraction and Inattention Detection," *IEEE Open Journal of the Industrial Electronics Society*, pp. 1–19, 2025, doi: 10.1109/OJIES.2025.3601982.
7. M. Fresta, F. Bellotti, I. Bochenko, L. Lazzaroni, G. Merlhiot, F. Tango, and R. Berta, "Deep Learning-Based Real-Time Driver Cognitive Distraction Detection," *IEEE Access*, vol. 13, pp. 26589–26607, 2025, doi: 10.1109/ACCESS.2025.3539392.
8. L. Lazzaroni, F. Bellotti, and R. Berta, "An embedded end-to-end voice assistant," *Engineering Applications of Artificial Intelligence*, vol. 136, p. 108998, Oct. 2024, doi: 10.1016/j.engappai.2024.108998.
9. F. Sakr, R. Berta, J. Doyle, A. Capello, A. Dabbous, L. Lazzaroni, and F. Bellotti, "CBin-NN: An Inference Engine for Binarized Neural Networks," *Electronics*, vol. 13, no. 9, Art. no. 9, Jan. 2024, doi: 10.3390/electronics13091624.
10. M. Fresta, F. Bellotti, A. Capello, A. Dabbous, L. Lazzaroni, F. Ansovini, and R. Berta, "End-to-End Dataset Collection System for Sport Activities," *Electronics*, vol. 13, no. 7, Art. no. 7, Jan. 2024, doi: 10.3390/electronics13071286.
11. R. Berta, L. Lazzaroni, A. Capello, M. Cossu, L. Forneris, A. Pighetti, and F. Bellotti, "Development of Deep-Learning-Based Autonomous Agents for Low-Speed Maneuvering in Unity," *Journal of Intelligent and Connected Vehicles*, vol. 7, no. 3, pp. 229–244, Sep. 2024, doi: 10.26599/JICV.2023.9210039.
12. R. Berta, A. Dabbous, L. Lazzaroni, D. Pau, and F. Bellotti, "Developing a TinyML Image Classifier in a Hour," *IEEE Open Journal of the Industrial Electronics Society*, pp. 1–15, 2024, doi: 10.1109/OJIES.2024.3451959.
13. A. Dabbous, R. Berta, M. Fresta, H. Ballout, L. Lazzaroni, and F. Bellotti, "Bringing Intelligence to the Edge for Structural Health Monitoring: The Case Study of the Z24 Bridge," *IEEE Open Journal of the Industrial Electronics Society*, vol. 5, pp. 781–794, 2024, doi: 10.1109/OJIES.2024.3434341.
14. L. Forneris, A. Pighetti, L. Lazzaroni, F. Bellotti, A. Capello, M. Cossu, and R. Berta, "Implementing Deep Reinforcement Learning (DRL)-based Driving Styles for Non-Player Vehicles," *International Journal of Serious Games*, vol. 10, no. 4, Art. no. 4, Nov. 2023, doi: 10.17083/ijsg.v10i4.638.

15. F. Ahmed, R. Berta, F. Bellotti, L. Lazzaroni, F. Floris, G. Barresi, and J. R. Carrión, “Socio-Economic Decision Making and Emotion Elicitation with a Serious Game in the Wild,” *Applied Sciences*, vol. 13, no. 11, Art. no. 11, Jan. 2023, doi: 10.3390/app13116432.
16. F. Bellotti, L. Lazzaroni, A. Capello, M. Cossu, A. De Gloria, and R. Berta, “Explaining a Deep Reinforcement Learning (DRL)-Based Automated Driving Agent in Highway Simulations,” *IEEE Access*, vol. 11, pp. 28522–28550, 2023, doi: 10.1109/ACCESS.2023.3259544.
17. R. Berta, F. Bellotti, A. De Gloria, and L. Lazzaroni, “Assessing Versatility of a Generic End-to-End Platform for IoT Ecosystem Applications,” *Sensors*, vol. 22, no. 3, Art. no. 3, Jan. 2022, doi: 10.3390/s22030713.

International Conferences Proceedings

1. L. Lazzaroni, F. Bellotti, A. Pighetti, A. Saad, O. Srouf, and R. Berta, “Embedded Deployment of an LLM-Based Voice Assistant,” in *Applications in Electronics Pervading Industry, Environment and Society*, M. Ruo Roch, A. Gómez, L. Lazzaroni, M. Martina, P. Motto Ros, N. Pazos Escudero, and R. Berta, Eds., Cham: Springer Nature Switzerland, 2026, pp. 273–278. doi: 10.1007/978-3-032-17174-0_39.
2. D. Faryar, R. Berta, M. Fresta, A. Saad, L. Lazzaroni, H. Ballout, O. Srouf, and F. Bellotti, “Lightweight Deep Learning for SOC Estimation of Various Lithium-Ion Batteries on Xilinx ZCU104 FPGA,” in *Applications in Electronics Pervading Industry, Environment and Society*, M. Ruo Roch, A. Gómez, L. Lazzaroni, M. Martina, P. Motto Ros, N. Pazos Escudero, and R. Berta, Eds., Cham: Springer Nature Switzerland, 2026, pp. 25–31. doi: 10.1007/978-3-032-17174-0_4.
3. S. G. Fashkhami, R. Berta, A. Pighetti, M. Fresta, L. Lazzaroni, H. Ballout, H. Rojhan, and F. Bellotti, “A Dual Branch Quantized Neural Network for Road Surface Classification Using Smartphone Sensors,” in *Applications in Electronics Pervading Industry, Environment and Society*, M. Ruo Roch, A. Gómez, L. Lazzaroni, M. Martina, P. Motto Ros, N. Pazos Escudero, and R. Berta, Eds., Cham: Springer Nature Switzerland, 2026, pp. 183–190. doi: 10.1007/978-3-032-17174-0_25.
4. O. L. Audi, F. Bellotti, R. Berta, Z. Liu, L. Lazzaroni, A. Pilla, H. Rojhan, and L. Forneris, “Improving Lane Change Performance in CARLA Through Curvature-Adaptive B-Spline Planning,” in *Applications in Electronics Pervading Industry, Environment and Society*, M. Ruo Roch, A. Gómez, L. Lazzaroni, M. Martina, P. Motto Ros, N. Pazos Escudero, and R. Berta, Eds., Cham: Springer Nature Switzerland, 2026, pp. 212–217. doi: 10.1007/978-3-032-17174-0_29.
5. V. Soltanmuradov, F. Bellotti, R. Berta, D. M. Gómez, and L. Lazzaroni, “Comparing Complete and Incomplete Formal Verification in Deep Learning-Based Object Detection,” in *Applications in Electronics Pervading Industry, Environment and Society*, M. Ruo Roch, A. Gómez, L. Lazzaroni, M. Martina, P. Motto Ros, N. Pazos Escudero, and R. Berta, Eds., Cham: Springer Nature Switzerland, 2026, pp. 286–291. doi: 10.1007/978-3-032-17174-0_41.
6. H. Ballout, R. Berta, L. Lazzaroni, F. Sakr, A. Pighetti, M. Fresta, and F. Bellotti, “Optimizing Binary AutoEncoders on STM32 Microcontrollers,” in *Applications in Electronics Pervading Industry, Environment and Society*, M. Ruo Roch, A. Gómez, L. Lazzaroni, M. Martina, P. Motto Ros, N. Pazos Escudero, and R. Berta, Eds., Cham: Springer Nature Switzerland, 2026, pp. 301–307. doi: 10.1007/978-3-032-17174-0_43.
7. H. Rojhan, F. Bellotti, L. Lazzaroni, M. Fresta, L. Forneris, A. Pighetti, and R. Berta, “Managing Performance Indicator (PI) Forms Through a State-of-the-Art Big Data Platform,” in *Applications in Electronics Pervading Industry, Environment and Society*, M. Ruo Roch, A. Gómez, L. Lazzaroni, M. Martina, P. Motto Ros, N. Pazos Escudero, and R. Berta, Eds., Cham: Springer Nature Switzerland, 2026, pp. 218–223. doi: 10.1007/978-3-032-17174-0_30.
8. A. Zare, F. Bellotti, R. Berta, L. Lazzaroni, L. Forneris, and A. Pighetti, “A CARLA Parking Environment for Training Deep-Reinforcement Learning Agents,” in *Applications in Electronics Pervading Industry, Environment and Society*, M. Ruo Roch, A. Gómez, L. Lazzaroni, M. Martina, P. Motto Ros, N. Pazos Escudero, and R. Berta, Eds., Cham: Springer Nature Switzerland, 2026, pp. 79–84. doi: 10.1007/978-3-032-17174-0_11.
9. A. Pighetti, F. Bellotti, S. Mohammady, L. Lazzaroni, H. Ballout, and R. Berta, “Exploring Neural Architecture Search (NAS) for Binary Neural Networks (BNNs) in the New WakeVision,” in *Applications in Electronics*

- Pervading Industry, Environment and Society, M. Ruo Roch, A. Gómez, L. Lazzaroni, M. Martina, P. Motto Ros, N. Pazos Escudero, and R. Berta, Eds., Cham: Springer Nature Switzerland, 2026, pp. 85–90. doi: 10.1007/978-3-032-17174-0_12.
10. A. Saad, F. Bellotti, H. Ballout, M. Fresta, L. Lazzaroni, O. Srouf, and R. Berta, “Resource-Adaptive FPGA-Based Kernels for Accelerating Self-attention in Transformers,” in *Applications in Electronics Pervading Industry, Environment and Society*, M. Ruo Roch, A. Gómez, L. Lazzaroni, M. Martina, P. Motto Ros, N. Pazos Escudero, and R. Berta, Eds., Cham: Springer Nature Switzerland, 2026, pp. 321–327. doi: 10.1007/978-3-032-17174-0_46.
 11. L. Motta, F. Bellotti, R. Berta, L. Lazzaroni, M. Fresta, and N. Busi, “Modeling Power Electronics in Simulink. A Tutorial for MSc Electronic Engineering Students,” in *Applications in Electronics Pervading Industry, Environment and Society*, M. Ruo Roch, A. Gómez, L. Lazzaroni, M. Martina, P. Motto Ros, N. Pazos Escudero, and R. Berta, Eds., Cham: Springer Nature Switzerland, 2026, pp. 242–248. doi: 10.1007/978-3-032-17174-0_34.
 12. E. Castagno, F. Bellotti, R. Berta, L. Lazzaroni, M. Fresta, and S. Incerti, “A Collaborative GitHub Framework for Project-Driven Learning in Microcontroller-Based Embedded Systems,” in *Applications in Electronics Pervading Industry, Environment and Society*, M. Ruo Roch, A. Gómez, L. Lazzaroni, M. Martina, P. Motto Ros, N. Pazos Escudero, and R. Berta, Eds., Cham: Springer Nature Switzerland, 2026, pp. 249–254. doi: 10.1007/978-3-032-17174-0_35.
 13. A. Dabbous, L. Lazzaroni, F. Sakr, R. Berta, M. Fresta, and F. Bellotti, “Binary Neural Networks for Real-Time Structural Health Monitoring on Edge Devices,” 2025 IEEE International Conference on Consumer Electronics (ICCE), Las Vegas, NV, USA, 2025, pp. 1–4, doi: 10.1109/ICCE63647.2025.10930047.
 14. A. Capello, R. Berta, M. Fresta, L. Lazzaroni, and F. Bellotti, “Leveraging Neural Architecture Search for Structural Health Monitoring on Resource-Constrained Devices,” in *Applications in Electronics Pervading Industry, Environment and Society*, M. Ruo Roch, F. Bellotti, R. Berta, M. Martina, and P. Motto Ros, Eds., Cham: Springer Nature Switzerland, 2025, pp. 323–330. doi: 10.1007/978-3-031-84100-2_38.
 15. A. Dabbous, R. Berta, L. Lazzaroni, A. Pighetti, and F. Bellotti, “Benchmarking Microcontrollers with Ultra-Low Resolution Images Classification,” in *Applications in Electronics Pervading Industry, Environment and Society*, M. Ruo Roch, F. Bellotti, R. Berta, M. Martina, and P. Motto Ros, Eds., Cham: Springer Nature Switzerland, 2025, pp. 234–241. doi: 10.1007/978-3-031-84100-2_28.
 16. L. Lazzaroni, A. Dabbous, M. Nardello, M. A. Lebdeh, R. Berta, K. S. Yildirim, F. Bellotti, and D. Brunelli, “Green Machine Learning for the IoT (GEMINI): A Position Paper,” in *Applications in Electronics Pervading Industry, Environment and Society*, M. Ruo Roch, F. Bellotti, R. Berta, M. Martina, and P. Motto Ros, Eds., Cham: Springer Nature Switzerland, 2025, pp. 339–346. doi: 10.1007/978-3-031-84100-2_40.
 17. M. Cossu, R. Berta, M. A. Bagheri Orumi, L. Lazzaroni, J.-L. Sauvaget, H. Touil, and F. Bellotti, “Comparing Detection-Based and Motion-Based Preprocessing of Video Frames for Automated Driving Scenario Detection,” in *Applications in Electronics Pervading Industry, Environment and Society*, M. Ruo Roch, F. Bellotti, R. Berta, M. Martina, and P. Motto Ros, Eds., Cham: Springer Nature Switzerland, 2025, pp. 250–257. doi: 10.1007/978-3-031-84100-2_30.
 18. F. Bellotti, R. Berta, V. Soltanmuradov, D. M. Gómez, A. Dhonthi, V. Hashemi, and L. Lazzaroni, “Robustness Verification of a Reinforcement Learning-Based Agent for Automated Car Parking,” in *Applications in Electronics Pervading Industry, Environment and Society*, M. Ruo Roch, F. Bellotti, R. Berta, M. Martina, and P. Motto Ros, Eds., Cham: Springer Nature Switzerland, 2025, pp. 139–147. doi: 10.1007/978-3-031-84100-2_17.
 19. L. Lazzaroni, F. Bellotti, A. Dabbous, A. Pighetti, and R. Berta, “Quantization of MobileNetV2 for Resource-Constrained Microcontrollers,” in *Applications in Electronics Pervading Industry, Environment and Society*, M. Ruo Roch, F. Bellotti, R. Berta, M. Martina, and P. Motto Ros, Eds., Cham: Springer Nature Switzerland, 2025, pp. 497–504. doi: 10.1007/978-3-031-84100-2_59.
 20. H. Ballout, R. Berta, I. Bochenko, L. Lazzaroni, A. Dhonthi, V. Hashemi, and F. Bellotti, “Runtime Input Monitoring of a Reinforcement Learning-Based Agent for Automated Car Parking,” in *Applications in*

- Electronics Pervading Industry, Environment and Society, M. Ruo Roch, F. Bellotti, R. Berta, M. Martina, and P. Motto Ros, Eds., Cham: Springer Nature Switzerland, 2025, pp. 387–394. doi: 10.1007/978-3-031-84100-2_46.
21. H. Ballout, R. Berta, A. Dabbous, M. Fresta, L. Lazzaroni, A. Pighetti, and F. Bellotti, “Performance Comparison of YOLOv8 and YOLOv10 for Traffic Light Detection on a Jetson Nano Board,” in *Applications in Electronics Pervading Industry, Environment and Society*, M. Ruo Roch, F. Bellotti, R. Berta, M. Martina, and P. Motto Ros, Eds., Cham: Springer Nature Switzerland, 2025, pp. 242–249. doi: 10.1007/978-3-031-84100-2_29.
 22. L. Forneris, F. Bellotti, R. Berta, M. Cossu, M. Fresta, R. Rezaieamale, H. Rojhan, V. Soltanmuradov, F. Tango, and L. Lazzaroni, “Setting Up a Lightweight Simulation Environment for Automated Driving Dataset Collection,” in *Applications in Electronics Pervading Industry, Environment and Society*, M. Ruo Roch, F. Bellotti, R. Berta, M. Martina, and P. Motto Ros, Eds., Cham: Springer Nature Switzerland, 2025, pp. 156–163. doi: 10.1007/978-3-031-84100-2_19.
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(Luca Lazzaroni)